1. An election result prediction is conducted by drawing a sample of 1000 people. Among these 1000 people, 600 actually voted for political party A and the remaining 400 voted for party B. Our prediction algorithm predicted that 400 people would vote for party A and the rest for party B, but among this predicted 400, only 300 people actually voted for party A. Prepare a confusion matrix. Compute accuracy and recall.
2. Construct a decision tree for the Boolean function A = (X AND Y) OR Z.
3. Assume you have a large dataset of features, but the number of distinct classes in the data is unknown. Suggest a suitable machine learning algorithm to group this data, and justify your answer.
4. An insurance company wants to identify groups of motor insurance policyholders with high average claim costs. Propose a suitable machine learning technique to solve this problem, with justification.
5. The following are the data points: A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9). Using the initial cluster centers c1(2,10), c2(5,8), and c3(1,2), determine how each data point is assigned to a cluster in the first iteration of the K-Means algorithm.